

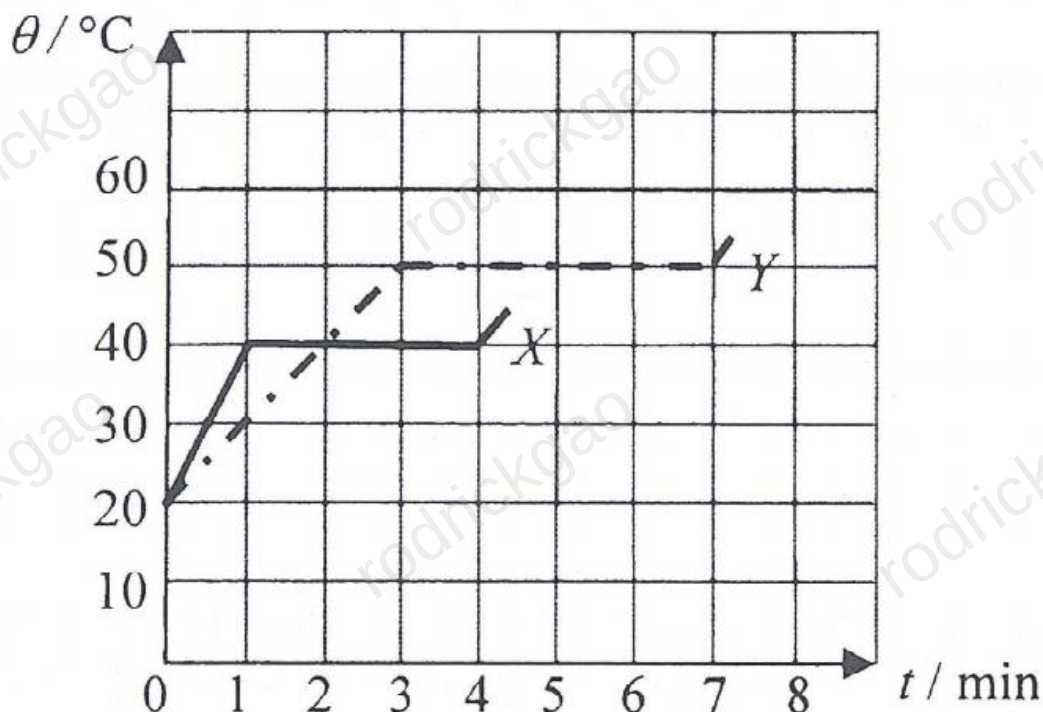
2021 HKDSE Physics

Paper 1A — Multiple-choice Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Question 1

Solid substances X and Y of equal mass are heated by heaters of power $2P$ and P respectively. The graph shows how the temperature θ of each substance varies with the heating time t . What is the ratio of the specific latent heat of fusion of X to that of Y?



- A 3 : 2
- B 3 : 4
- C 4 : 3
- D 2 : 3

Official answer: A

During melting, $Q = Pt = ml$. From the plateaux in the graph, $t_X:t_Y = 3:4$; since the masses are equal and the heater powers are $2P$ and P , $l_X:l_Y = 2t_X:t_Y = 6:4 = 3:2$. Answer: A.

Question 2

Metal blocks X and Y are identical in size and shape. X is of a higher temperature than Y. Which of the following statements must be correct?

- (1) Energy will flow from X to Y if they are in thermal contact.
- (2) X is a better conductor of heat compared to Y.
- (3) The total internal energy of X is higher than that of Y.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

Official answer: A

Heat flows spontaneously from the body at higher temperature to the body at lower temperature when they are in thermal contact, so statement (1) must be true. Temperature alone does not determine thermal conductivity or total internal energy. Answer: A.

Question 3*

For an ideal gas, kinetic theory deduces that $pV = (1/3)Nmc^2$. Which physical quantity below can be represented by $3p/c^2$?

A total mass of gas

B volume of one mole

C density

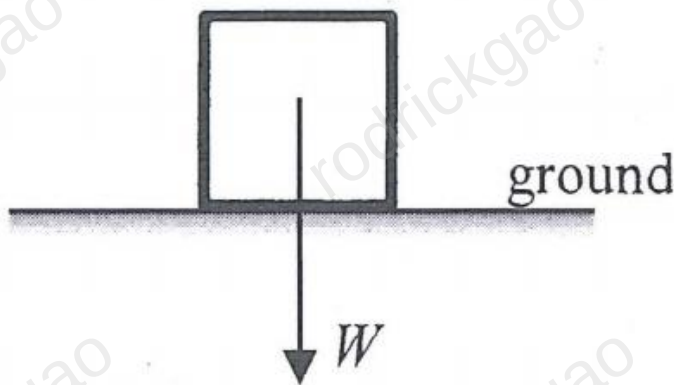
D number of gas molecules per unit volume

Official answer: C

From $pV = (1/3)Nmc^2$, $3p/c^2 = Nm/V$. Since Nm is the total gas mass, Nm/V is the gas density. Answer: C.

Question 4

A block of weight W is at rest on a horizontal ground as shown. The force acting on the block by the ground is R . Which of the following statements are correct?



(1) R and W are opposite in direction.

(2) R and W are equal in magnitude.

(3) R and W form an action–reaction pair.

A (1) only

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

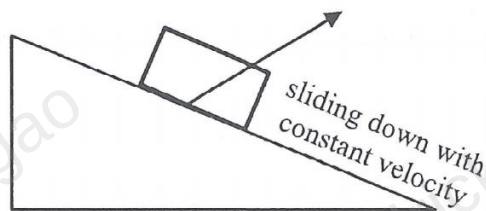
Official answer: B

The block is at rest, so the ground force R balances the weight W : they are equal in magnitude and opposite in direction. They act on the same body and therefore are not an action–reaction pair. Thus (1) and (2) are correct.
Answer: B.

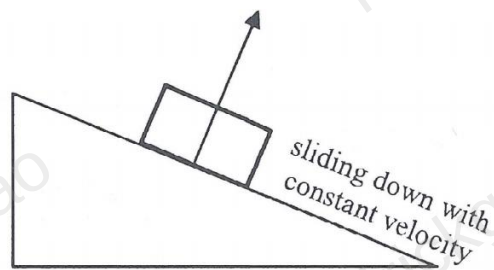
Question 5

A block is sliding down a rough incline with constant velocity as shown. Which arrow indicates the direction of the resultant force acting on the block by the incline? Neglect air resistance.

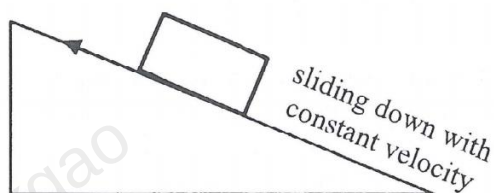
A.



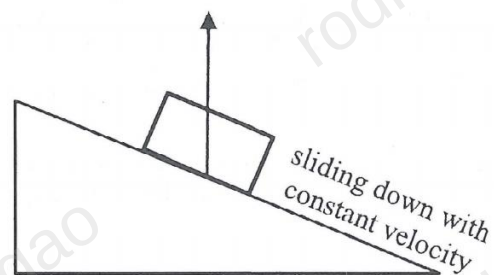
B.



C.



D.



A Arrow A in the diagram

B Arrow B in the diagram

C Arrow C in the diagram

D Arrow D in the diagram

Official answer: D

Constant velocity means zero resultant force. The single resultant contact force exerted by the incline must therefore be vertically upward to balance the weight; this is arrow D in the diagram. Answer: D.

Question 6

Which statement below about a raindrop falling at a constant terminal velocity is correct?



raindrop falling at
a constant velocity

- A No work is done by the gravitational force.
- B All the loss in gravitational potential energy is converted to a gain in kinetic energy.
- C The only force acting on the raindrop is its weight.
- D No net force acts on the raindrop.

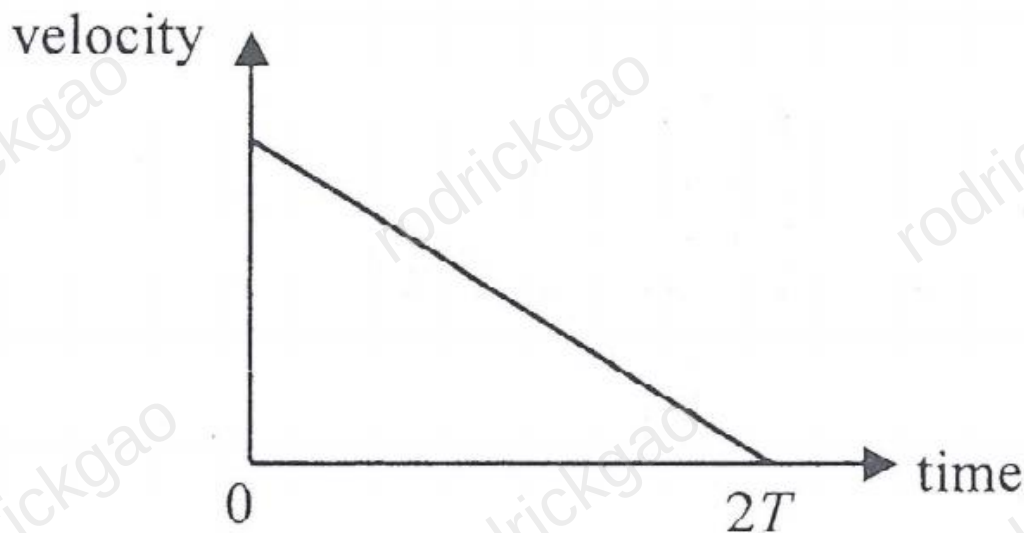
Official answer: D

At terminal velocity the upward resistive forces exactly balance the weight, so the resultant force is zero.

Gravitational potential energy is continually dissipated rather than converted into kinetic energy. Answer: D.

Question 7

At time $t = 0$, a small sphere is projected up along a smooth incline with a certain initial velocity. It travels a distance L and becomes momentarily at rest after a time $2T$. The corresponding velocity–time graph is shown. What is the distance travelled by the sphere from $t = 0$ to $t = T$?



- A $L/4$
- B $L/2$
- C $3L/4$**
- D $4L/5$

Official answer: C

The velocity falls uniformly from u at $t = 0$ to zero at $2T$, so at T it is $u/2$. The distance in the first T is $[(u + u/2)/2]T = 3uT/4$, while the total distance L is $(1/2)(2T)u = uT$. Hence the required distance is $3L/4$. Answer: C.

Question 8

A stationary uranium nucleus undergoes the decay $^{238}_{92}\text{U} \rightarrow ^{234}_{90}\text{Th} + ^4_2\text{He}$. Which statement about the thorium-234 nucleus and the α particle just after the decay is correct?

- A $p(\text{Th}) = p(\alpha)$, and $\text{KE}(\text{Th}) < \text{KE}(\alpha)$
 - B $p(\text{Th}) > p(\alpha)$, and $\text{KE}(\text{Th}) > \text{KE}(\alpha)$
 - C $p(\text{Th}) = p(\alpha)$, and $\text{KE}(\text{Th}) > \text{KE}(\alpha)$
 - D $p(\text{Th}) = p(\alpha)$, and $\text{KE}(\text{Th}) = \text{KE}(\alpha)$
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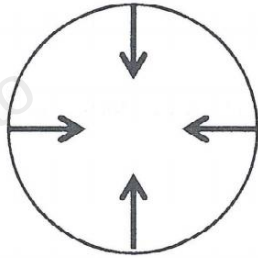
Official answer: A

The original nucleus is stationary, so conservation of momentum gives equal and opposite momenta for Th-234 and the α -particle. For the same momentum, $K = p^2/(2m)$, so the much heavier thorium nucleus has less kinetic energy. Answer: A.

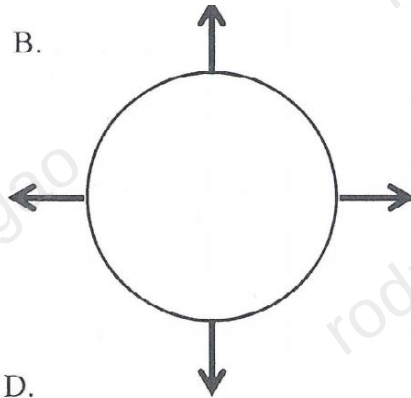
Question 9*

A particle travels clockwise in a uniform circular path. Which top-view diagram correctly shows the net force acting on the particle at the instant shown?

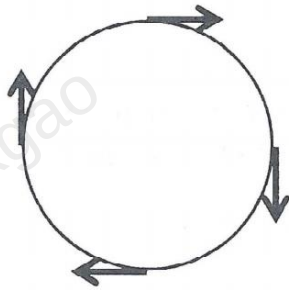
A.



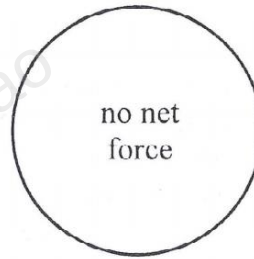
B.



C.



D.



A Diagram A

B Diagram B

C Diagram C

D Diagram D

Official answer: A

Uniform circular motion requires a resultant force directed towards the centre of the circle. At the shown position this is the direction indicated in diagram A. Answer: A.

Question 10

Mechanical waves can transfer which of the following?

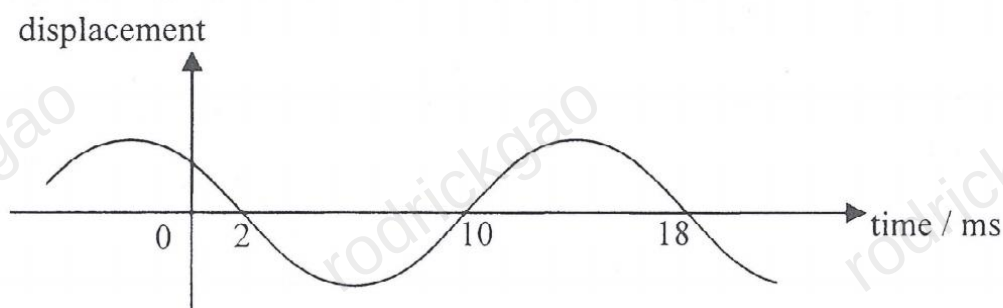
- (1) mass
 - (2) momentum
 - (3) energy
- A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

Official answer: C

Mechanical waves transfer energy and can transfer momentum, but the particles of the medium only oscillate about equilibrium positions, so there is no net transfer of mass. Thus (2) and (3) are correct. Answer: C.

Question 11

The displacement–time graph of a particle in a travelling wave is shown. What is the frequency of the wave?



- A 55.6 Hz
B 62.5 Hz
C 111 Hz
D 125 Hz

Official answer: B

One complete cycle on the displacement–time graph takes 0.016 s. Hence $f = 1/T = 1/0.016 = 62.5$ Hz. Answer: B.

Question 12

P-waves and S-waves travel at 9.6 km s^{-1} and 6.4 km s^{-1} respectively. The P-waves arrive at a station at 7:02 a.m. and the S-waves at 7:04 a.m. Estimate the time when the earthquake occurred.

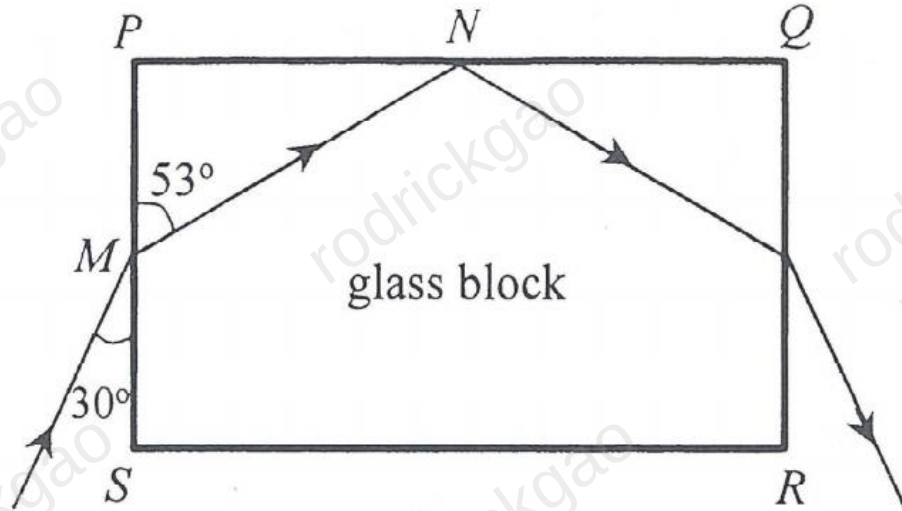
- A 6:53 a.m.
 - B 6:56 a.m.
 - C 6:58 a.m.
 - D 6:59 a.m.
-

Official answer: C

Let the distance be d . The 120 s arrival-time difference gives $d/6400 - d/9600 = 120$, so $d = 2.304 \times 10^6 \text{ m}$. The P-wave travel time is $d/9600 = 240 \text{ s} = 4 \text{ min}$; the earthquake occurred at 6:58 a.m. Answer: C.

Question 13

A light ray enters a rectangular glass block PQRS at M on face PS and then strikes face PQ at N, as shown. What is the critical angle for the glass–air boundary?



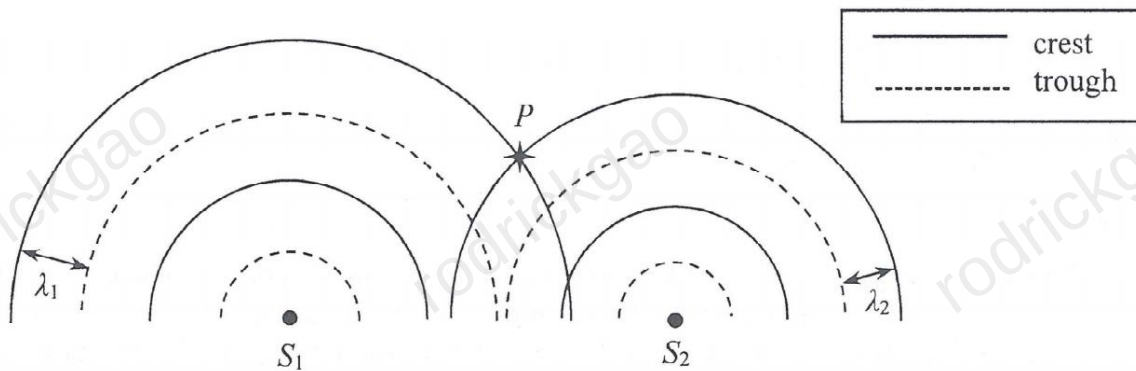
- A 37°
- B 44°**
- C 53°
- D 60°

Official answer: B

Applying Snell's law to the ray at M gives the refractive index of the glass from the shown angles. Then $\sin c = 1/n$ gives $c \approx 44^\circ$. Answer: B.

Question 14

Two vibrators S_1 and S_2 generate circular water waves of wavelengths λ_1 and λ_2 respectively, where $\lambda_1 > \lambda_2$. Which statement about point P is correct?



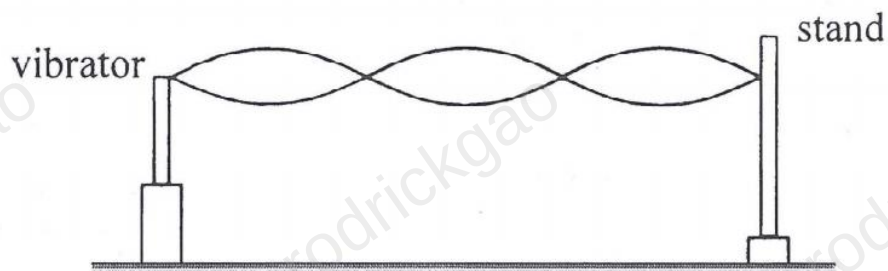
- A P is always a crest.
- B The waves from S_1 and S_2 always reinforce at P.
- C The principle of superposition cannot be applied because $\lambda_1 \neq \lambda_2$.
- D The principle of superposition applies, but the waves do not always reinforce at P.

Official answer: D

Superposition applies to waves even when their wavelengths differ. Because their phase difference at P changes with time, they do not always reinforce one another. Answer: D.

Question 15

A stationary wave is formed on a stretched string by a vibrator. The frequency of the vibrator is increased. Which changes may occur?



- (1) The number of antinodes increases.
- (2) The wave speed along the string increases.
- (3) The frequency of the sound produced in air increases.

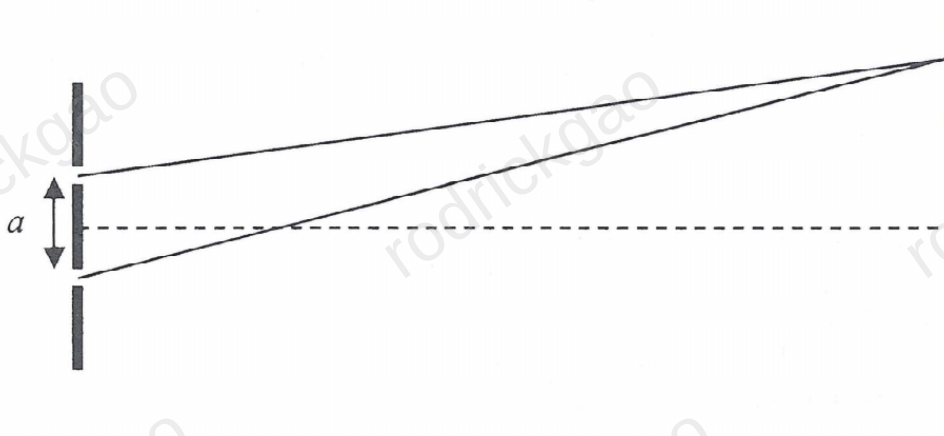
- A (1) and (2) only
- B (1) and (3) only**
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: B

For the same stretched string, $v = \sqrt{T/\mu}$ is unchanged. A higher driving frequency can excite a mode with more antinodes, and the sound produced in air has the higher frequency. Thus (1) and (3) are correct. Answer: B.

Question 16

In a Young double-slit experiment, the slit separation a is reduced while all other conditions remain unchanged. Which statements are correct?



- (1) The fringes become brighter.
 (2) The number of fringes observed increases.
 (3) The fringe separation becomes larger.

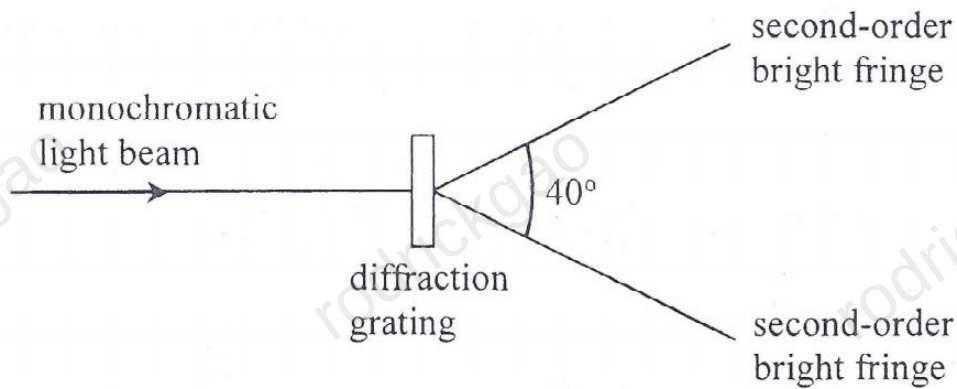
- A (1) only
 B (3) only
 C (1) and (2) only
 D (2) and (3) only

Official answer: B

The fringe separation is $\beta = \lambda D/a$. Reducing slit separation a therefore increases the fringe separation. It does not make the fringes brighter, and fewer—not more—fringes fit across a fixed field of view. Only statement (3) is correct. Answer: B.

Question 17*

A diffraction grating has 300 lines per millimetre. The angle between the two second-order fringes is 40° . What is the frequency of the light?



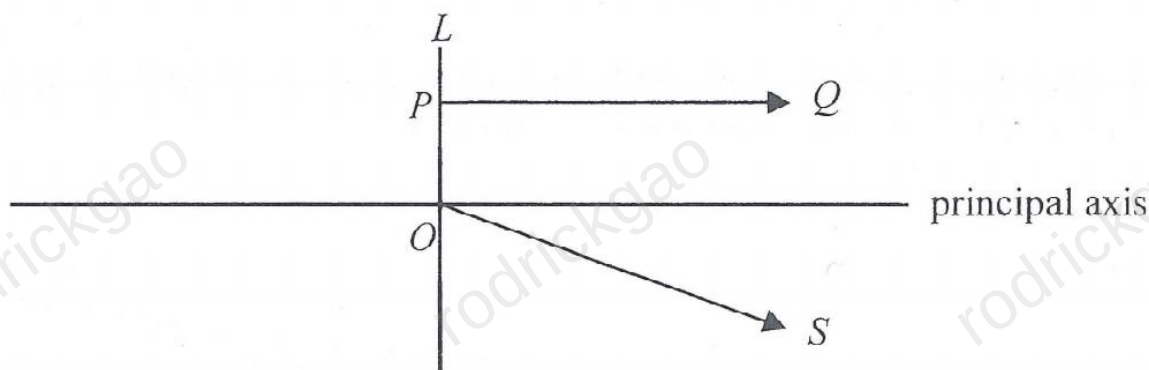
- A 1.4×10^{14} Hz
- B 2.6×10^{14} Hz
- C 2.8×10^{14} Hz
- D 5.3×10^{14} Hz

Official answer: D

The grating spacing is $d = 1/(300 \text{ mm}^{-1}) = 3.33 \times 10^{-6} \text{ m}$. The second-order fringes are at $\pm 20^\circ$, so $2\lambda = d \sin 20^\circ$ and $\lambda = 5.70 \times 10^{-7} \text{ m}$. Thus $f = c/\lambda = 5.26 \times 10^{14} \text{ Hz}$. Answer: D.

Question 18

Light rays PQ and QS are refracted by a lens L as shown. A point object is placed to the left of L. Which statements are correct?



- (1) The image formed must be virtual.
- (2) The object lies along the straight line containing QS.
- (3) L is a concave lens.

- A (1) only
- B (3) only
- C (1) and (2) only**
- D (2) and (3) only

Official answer: C

Tracing the refracted rays and their backward extensions shows that the image is virtual and that the object lies on the straight line containing QS. The lens need not be concave. Thus (1) and (2) are correct. Answer: C.

Question 19

The ratio of the typical wavelength of X-rays to that of microwaves is 10^n : 1. Which value could n be?

- A -10**
- B -4
- C +4
- D +10

Official answer: A

Typical X-ray wavelengths are many orders of magnitude shorter than microwave wavelengths; a ratio of 10^{-10} :1 is physically possible within their wavelength ranges. Answer: A.

Question 20

Submarines use ultrasound instead of microwaves for underwater communication because

- A ultrasound has a shorter wavelength.
 - B ultrasound travels faster in sea water.
 - C microwaves are easily absorbed by sea water.
 - D microwaves diffract too much.
-

Official answer: C

Sea water strongly absorbs microwaves, whereas ultrasound propagates through it and can be detected after reflection. The decisive reason is the strong microwave absorption. Answer: C.

Question 21

Three identical conducting spheres X, Y and Z carry charges $+2Q$, $-4Q$ and $+5Q$ respectively. Y is first brought into contact with X and then separated; Y is then brought into contact with Z and separated. What are the final charges?

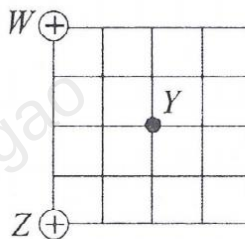
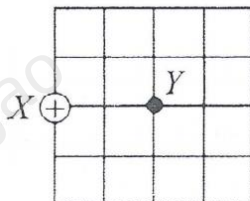
- A $X = 0$, $Y = +1.5Q$, $Z = +1.5Q$
 - B $X = -Q$, $Y = +0.5Q$, $Z = +0.5Q$
 - C $X = +Q$, $Y = +Q$, $Z = +Q$
 - D $X = -Q$, $Y = +2Q$, $Z = +2Q$
-

Official answer: D

After X touches Y, their total charge $-2Q$ is shared equally, so $X = Y = -Q$. Then Y and Z have total charge $4Q$ and share it equally, giving $Y = Z = +2Q$. Answer: D.

Question 22*

A point charge $+Q$ at X produces an electric field E_0 at Y. If identical charges $+Q$ are also placed at W and Z as shown, what is the resultant electric field at Y? Take $\sin 45^\circ = \cos 45^\circ = \sqrt{2}/2$.



A $(\sqrt{2}/2)E_0$

B E_0

C $\sqrt{2} E_0$

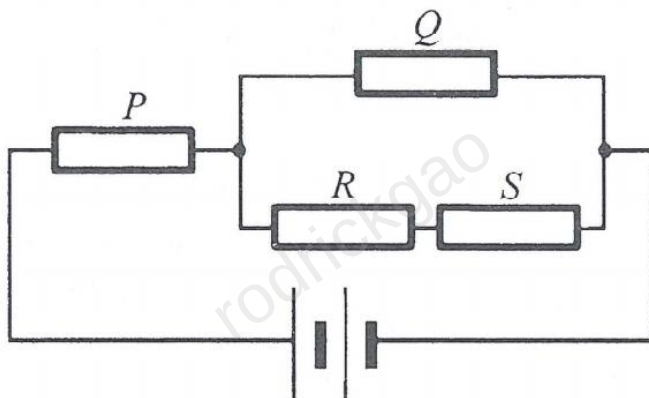
D $2E_0$

Official answer: A

At Y, each charge at W or Z is $2\sqrt{2}$ grid units away, compared with 2 units for X, so each produces $E_0/2$. Their vertical components cancel and their horizontal components add: $E = 2(E_0/2)\cos 45^\circ = E_0/\sqrt{2}$. Answer: A.

Question 23

Four identical resistors P, Q, R and S are connected to a battery as shown. If the power dissipated by R is 1 W, what is the total power supplied by the battery?



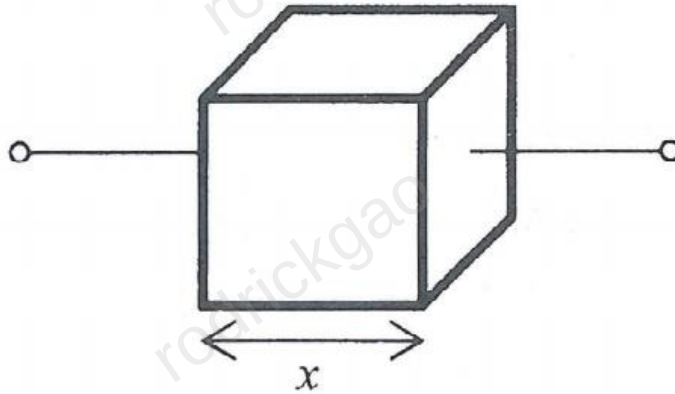
- A 11 W
- B 15 W**
- C 19 W
- D 21 W

Official answer: B

Let the current through R and S be I . Since each has power $I^2R = 1$ W, the parallel branch voltage is $2IR$. Hence Q carries $2I$ and dissipates 4 W; P carries $3I$ and dissipates 9 W. Total power = $9 + 4 + 1 + 1 = 15$ W. Answer: B.

Question 24

A metallic cube of side x has resistance R measured across two opposite faces. Which relation is correct?



A $R \propto 1/x$

B $R \propto x$

C $R \propto x^2$

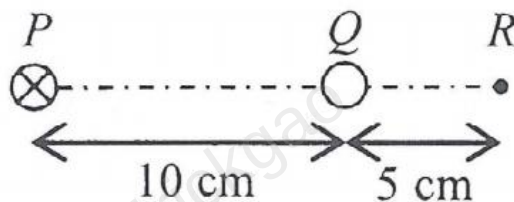
D $R \propto 1/x^2$

Official answer: A

For opposite faces of a cube, $R = \rho L/A = \rho x/x^2 = \rho/x$. Therefore $R \propto 1/x$. Answer: A.

Question 25

P, Q and R lie on a straight line with $PQ = 10\text{ cm}$ and $QR = 5\text{ cm}$. A long wire at P carries 0.3 A into the paper. A wire at Q carries current I so that the magnetic field at R is zero. Find the direction and magnitude of I .



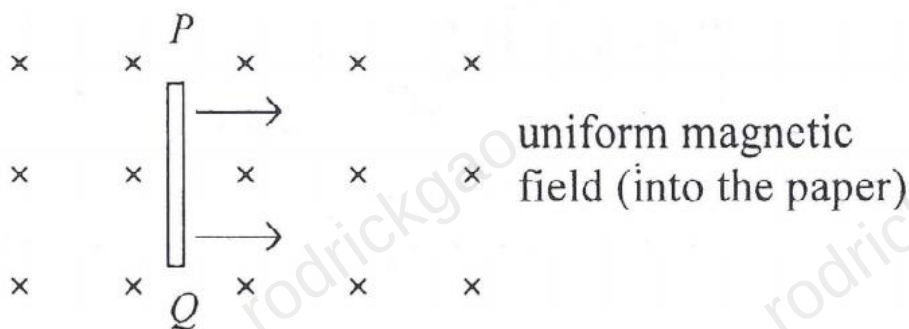
- A into the paper, 0.1 A
- B into the paper, 0.9 A
- C out of the paper, 0.1 A**
- D out of the paper, 0.9 A

Official answer: C

At R, the field from the 0.3 A current at P must be cancelled by the field from Q, so the Q current is out of the paper. Using $B = \mu_0 I / (2\pi r)$, $I/0.05 = 0.3/0.15$, giving $I = 0.10\text{ A}$. Answer: C.

Question 26

A copper rod PQ moves with constant velocity to the right across a uniform magnetic field directed into the paper. Which statements about the induced e.m.f. are correct?



- (1) Its magnitude depends on the length of the rod.
- (2) P is at a higher potential than Q.
- (3) The magnetic force on the rod opposes its motion.

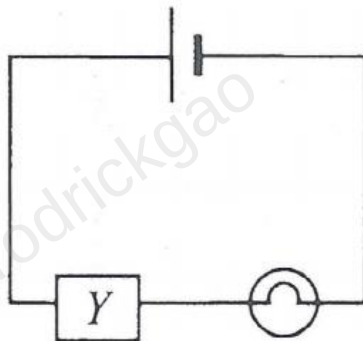
- A (1) only
- B (3) only
- C (1) and (2) only**
- D (2) and (3) only

Official answer: C

The motional e.m.f. $\varepsilon = Blv$ depends on the rod length. From $q(\mathbf{v} \times \mathbf{B})$, positive charge is driven towards P, so P is at higher potential. With no closed circuit there is no sustained induced current and hence no magnetic braking force. Thus (1) and (2) are correct. Answer: C.

Question 27

A light bulb and a device Y are connected in series with an ideal cell. The brightness of the bulb decreases. Which statements must be correct?



- (1) The current through Y decreases.
- (2) The voltage across Y decreases.
- (3) The power supplied by the cell decreases.

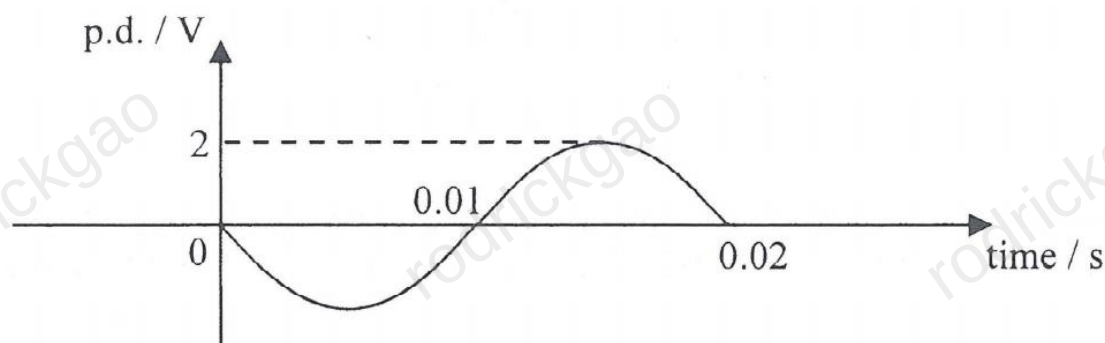
- A (1) and (2) only
- B (1) and (3) only**
- C (2) and (3) only
- D (1), (2) and (3)

Official answer: B

The dimmer series bulb means the circuit current decreases, so the current through Y decreases and the cell power EI also decreases. The bulb voltage falls, so the voltage across Y need not decrease. Thus (1) and (3) are correct. Answer: B.

Question 28*

A sinusoidal alternating p.d. is applied across a $10\ \Omega$ resistor, as shown. Find the r.m.s. current and the average power dissipated.



- A 0.14 A; 0.2 W
- B 0.14 A; 0.4 W
- C 0.20 A; 0.2 W
- D 0.20 A; 0.4 W

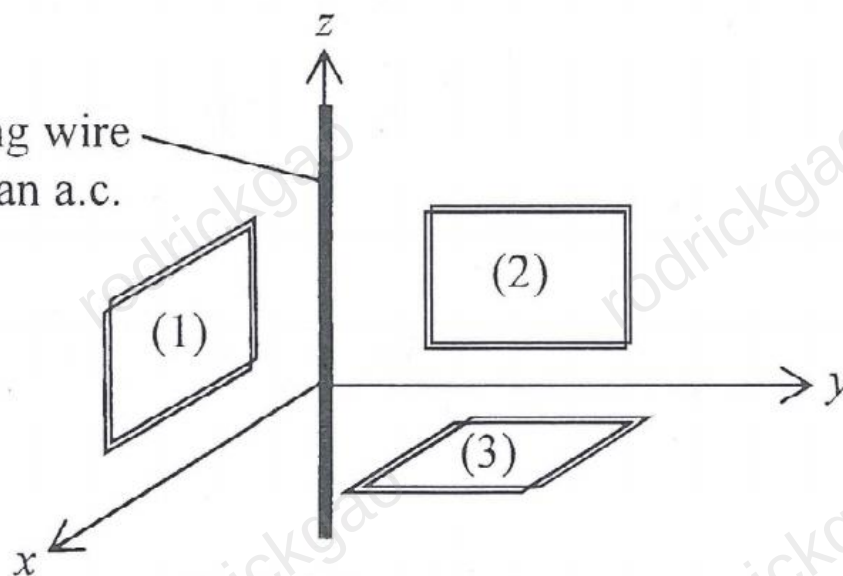
Official answer: A

From the graph $V_{\text{peak}} = 2.0\ \text{V}$. Therefore $I_{\text{rms}} = V_{\text{peak}}/(\sqrt{2}R) = 2/(\sqrt{2} \times 10) = 0.14\ \text{A}$, and $P_{\text{av}} = I_{\text{rms}}^2 R = 0.20\ \text{W}$.

Answer: A.

Question 29

Three mutually perpendicular coils are placed near a straight conducting wire along the z -axis carrying an alternating current. In which coils is an e.m.f. induced?



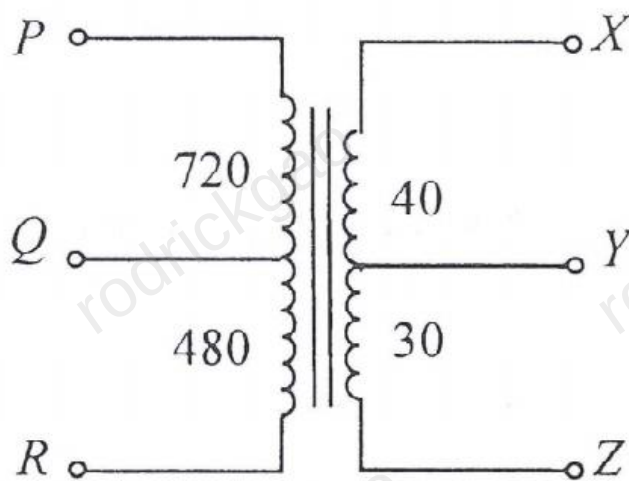
- A (1) only
- B (3) only
- C (1) and (2) only
- D (2) and (3) only

Official answer: C

An e.m.f. is induced only when the alternating magnetic flux links the coil. From the orientations shown, coils (1) and (2) have changing flux through them, whereas coil (3) does not. Answer: C.

Question 30*

A multi-tapped transformer is used to step 240 V down to 6 V. Which pair of terminals should be used for the primary and secondary coils?



- A primary PQ; secondary XY
- B primary PQ; secondary YZ
- C primary PR; secondary XY
- D primary PR; secondary YZ**

Official answer: D

For the largest step-down ratio, use the greatest available number of primary turns and the smallest available number of secondary turns. The correct terminals are PR for the primary and YZ for the secondary. Answer: D.

Question 31

A radioactive nuclide plutonium-239 ($^{239}_{94}\text{Pu}$) becomes a stable lead-207 isotope ($^{207}_{82}\text{Pb}$) after a series of α and β decays. Find the number of β decays in the process.

- A 3
- B 4**
- C 5
- D 6

Official answer: B

The nucleon number falls from 239 to 207, requiring eight α -decays. These reduce the proton number from 94 to 78; four β -decays then raise it to 82. Answer: B.

Question 32

The activity of a radioactive sample is 18 MBq. What was its activity three half-lives earlier?

- A 6 MBq
 - B 54 MBq
 - C 72 MBq
 - D 144 MBq
-

Official answer: D

Three half-lives earlier the activity was larger by 2^3 . Hence $A_0 = 18 \times 8 = 144$ MBq. Answer: D.

Question 33

Which of the following may contain sources of ionizing radiation?

- (1) sea water
- (2) rock
- (3) the human body

- A (1) only
 - B (2) only
 - C (2) and (3) only
 - D (1), (2) and (3)
-

Official answer: D

Sea water contains natural radioactive isotopes, rocks contain terrestrial radionuclides, and the human body contains isotopes such as K-40 and C-14. All three may therefore contain ionizing-radiation sources. Answer: D.